

The Grothendieck-type exponent conjecture is already answered

Literature status for arXiv:1102.4542

13 August 2026

Status: literature already answered. Pellegrino–Seoane-Sepúlveda, arXiv:1307.4809, explicitly identifies and solves Bernardino’s 2011 conjecture.

Source question

A. T. Bernardino, *On cotype and a Grothendieck-type theorem for absolutely summing multilinear operators*, arXiv:1102.4542; Quaest. Math. **34** (2011), 513–519.

Problem 1.3 on PDF page 2 asks for the best $g_n > 0$ such that

$$\Pi_{(g_n; 1, \dots, 1)}^n(\ell_1, \dots, \ell_1; \ell_2) = \mathcal{L}(\ell_1, \dots, \ell_1; \ell_2).$$

The final theorem proves $g_n \leq 2/(n+1)$, and the last paragraph conjectures that $2/(n+1)$ is optimal.

Exact later answer

D. Pellegrino and J. B. Seoane-Sepúlveda, *Grothendieck’s theorem for absolutely summing multilinear operators is optimal*, arXiv:1307.4809; Linear Multilinear Algebra **63** (2015), 554–558, DOI 10.1080/03081087.2013.877013.

Its abstract states that the result “solves (in the positive) a conjecture posed by A. T. Bernardino in 2011.” Theorem 1.1 on PDF page 1 proves

$$g_n = \frac{2}{n+1}.$$

More strongly, for every $r < 2/(n+1)$ there is a continuum-dimensional linear space whose nonzero elements are continuous n -linear maps $\ell_1^n \rightarrow \ell_2$ that are not absolutely $(r; 1, \dots, 1)$ -summing.

Identification

The notation and domain/range are identical after renaming the arity m in the supporting paper to n in the source. The source supplies the upper bound, while Theorem 1.1 supplies failure for every smaller exponent. Thus the source’s sharp-exponent problem and final conjecture are fully resolved. The source’s broader Problem 1.5 about optimal cotype-dependent inclusion exponents is a separate question and is not claimed here.